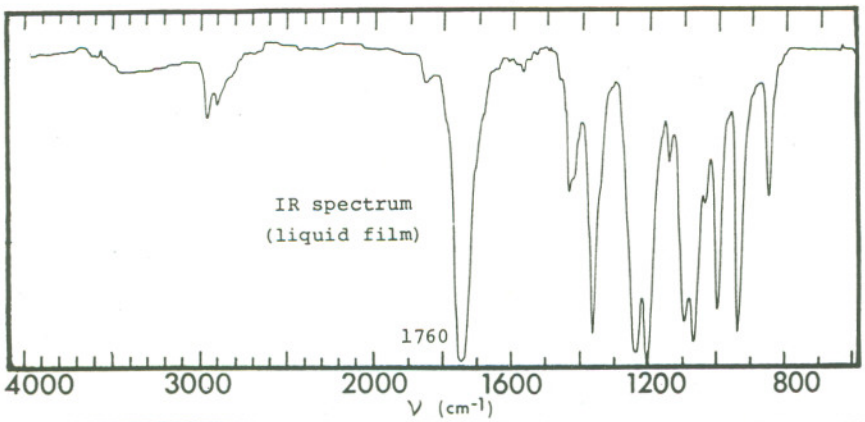
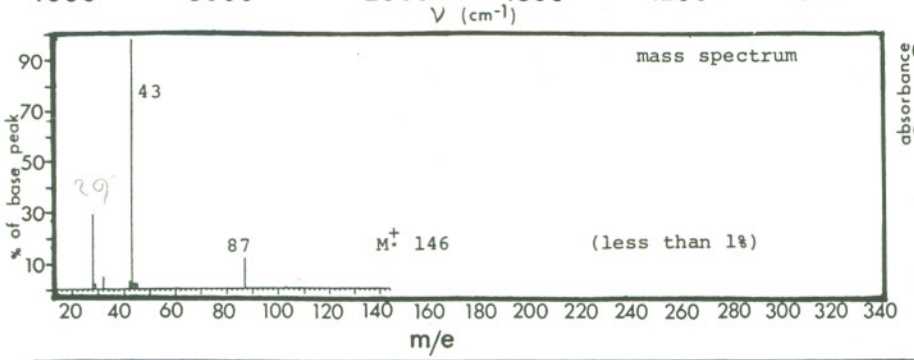
Organic Cume

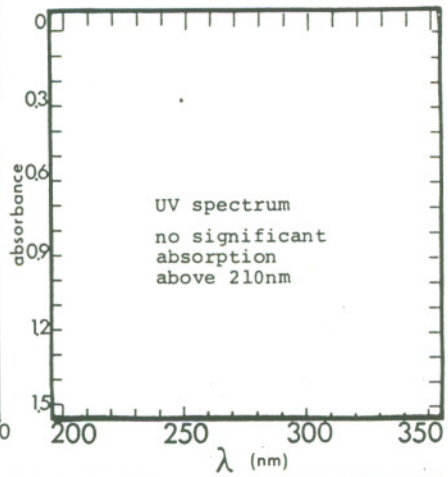
January 16, 1993

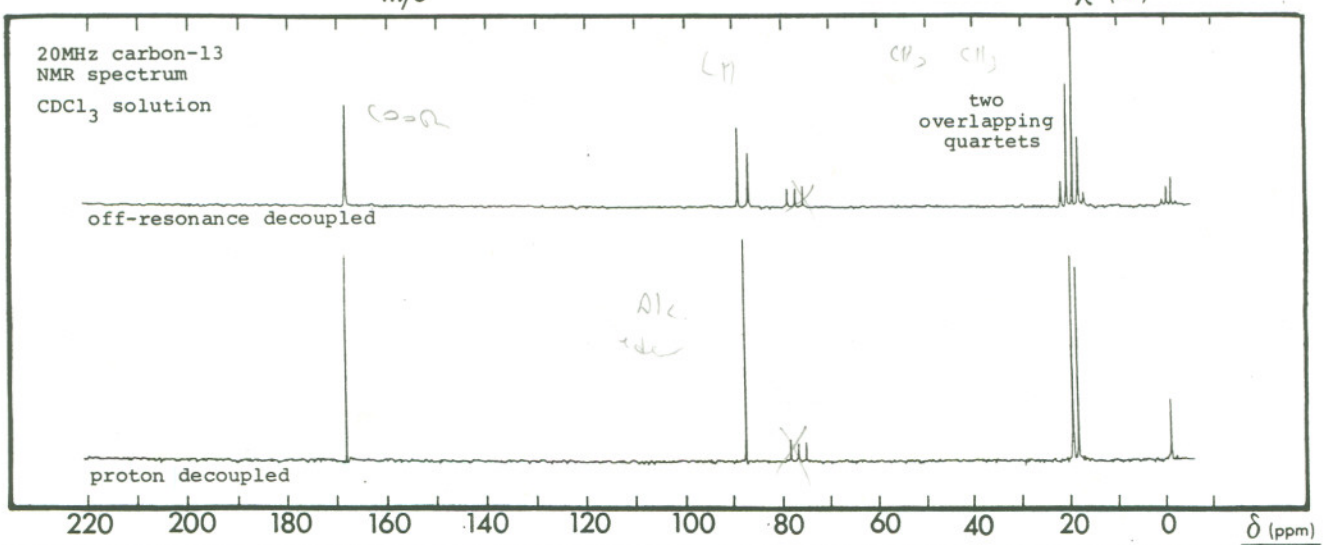
Problem 1

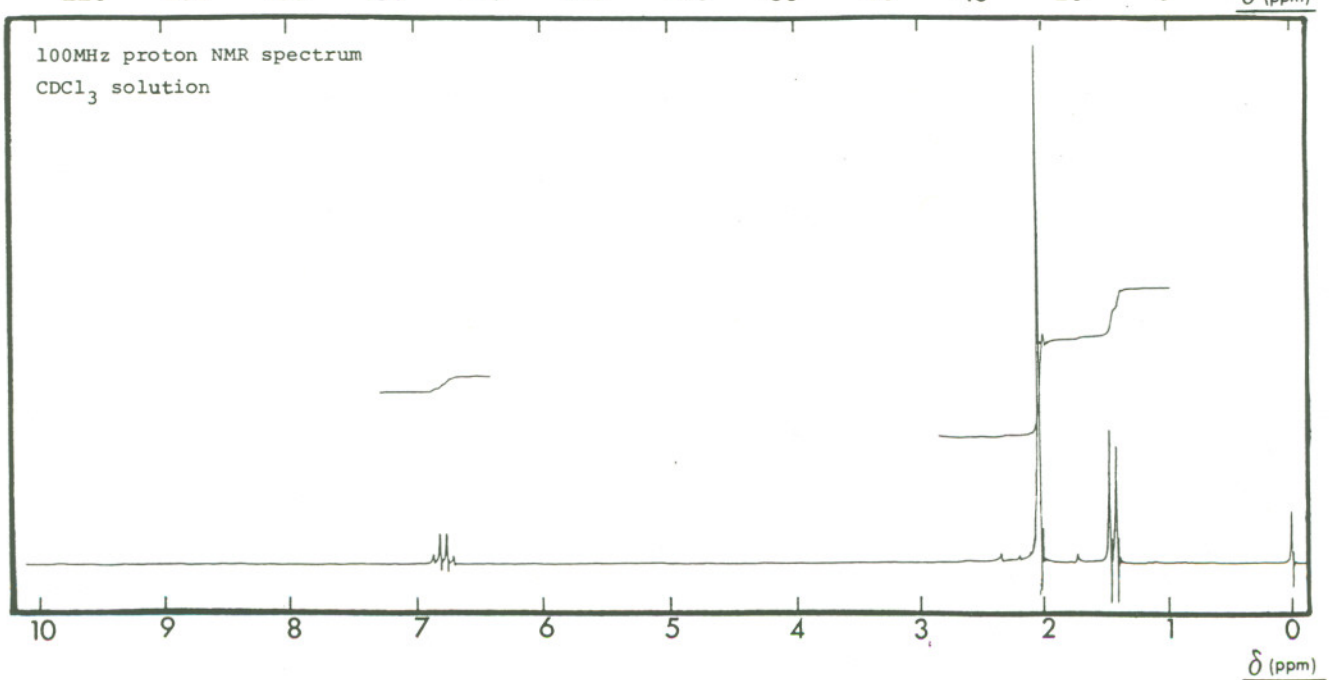
Give a structure which corresponds to each problem.



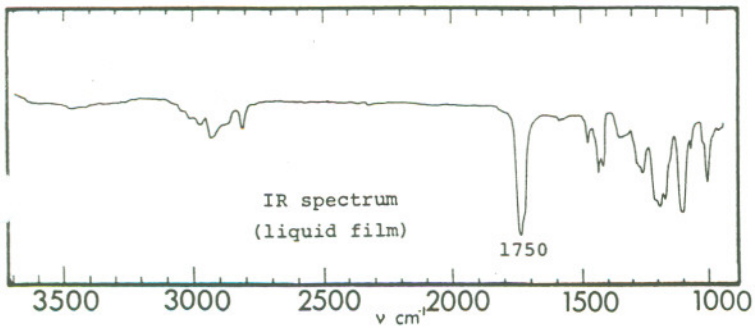


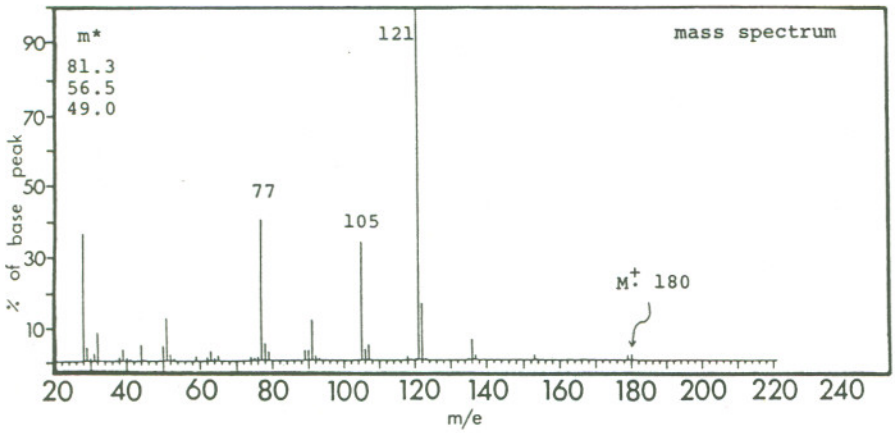


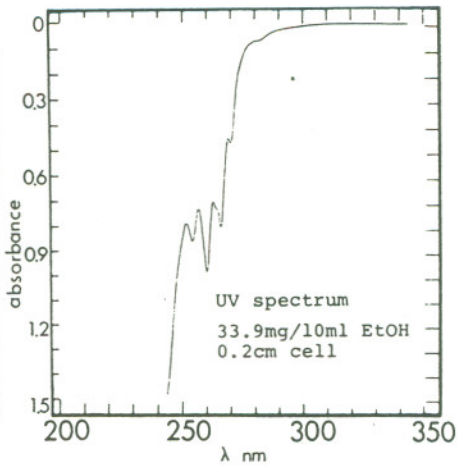


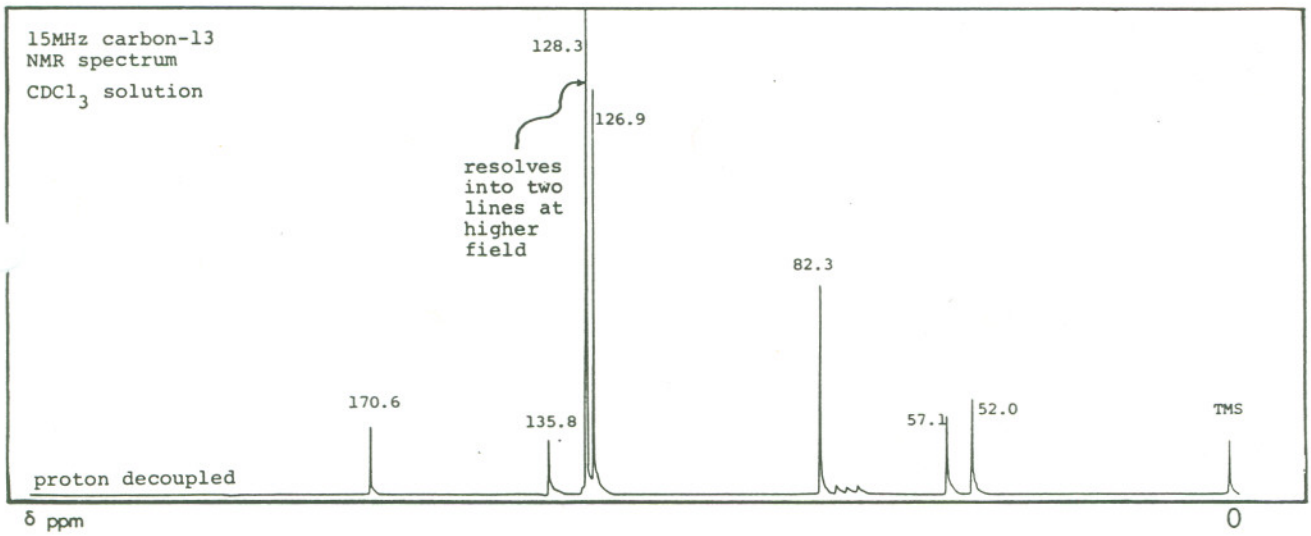


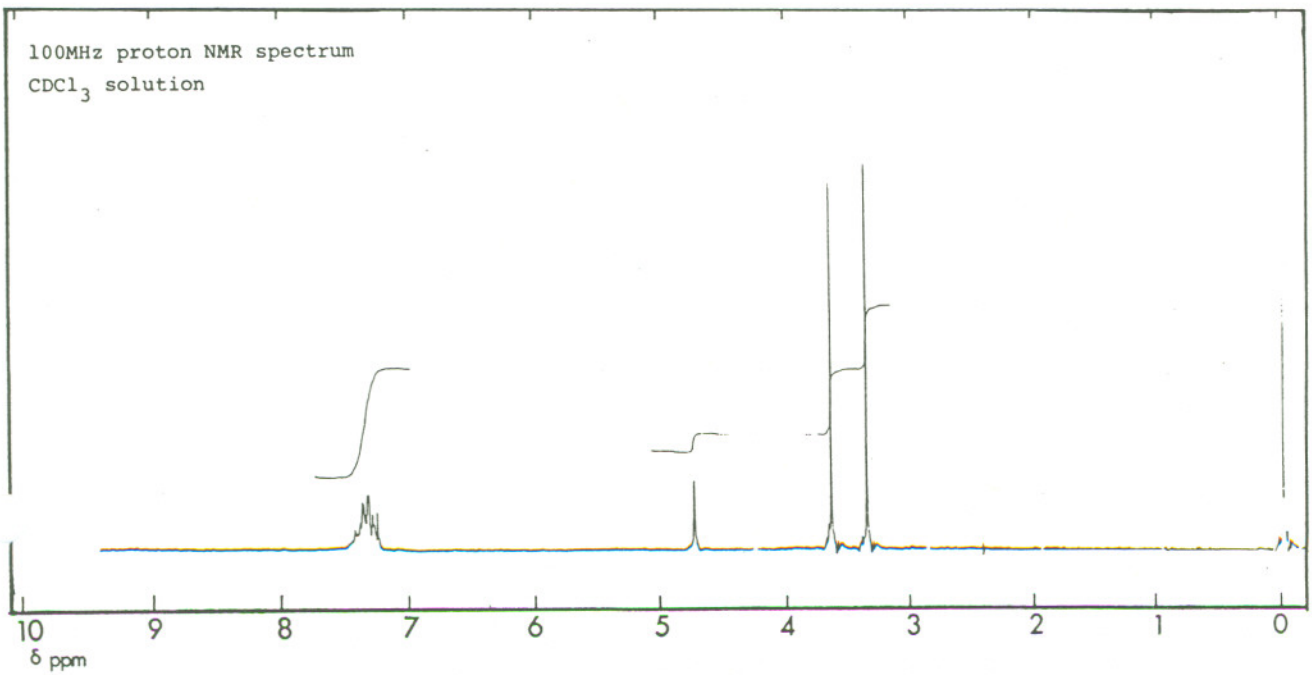
Problem 2


IR spectrum
(liquid film)
1750
3500
3000
2500
2000
1500
1000
ν cm⁻¹

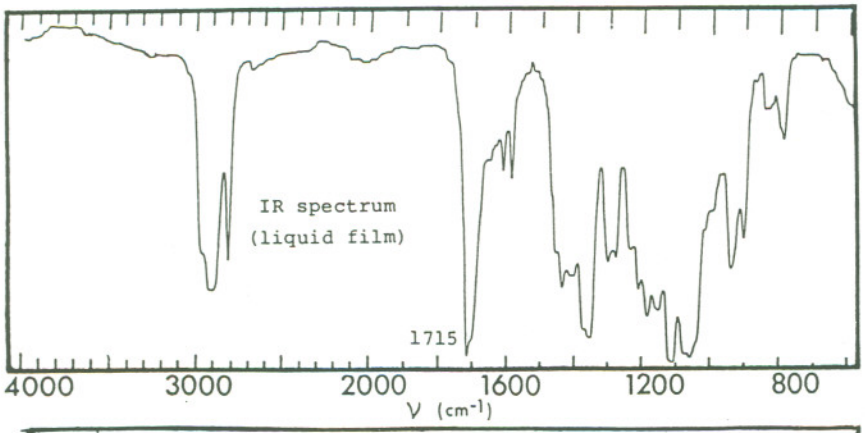

m*
81.3
56.5
49.0
121
mass spectrum
77
105
M.+ 180
% of base peak
90
70
50
30
10
20
40
60
80
100
120
140
160
180
200
220
240
m/e

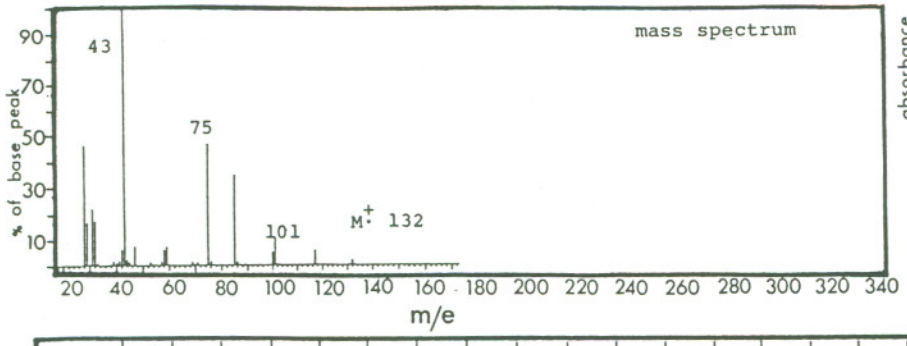

absorbance
0
0.3
0.6
0.9
1.2
1.5
UV spectrum
33.9mg/10ml EtOH
0.2cm cell
200
250
300
350
λ nm

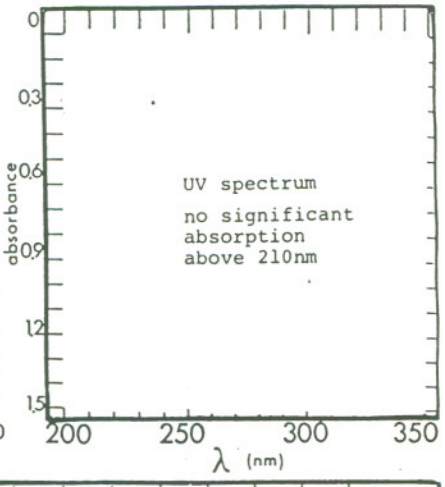

15MHz carbon-13
NMR spectrum
CDCl3 solution
128.3
126.9
resolves
into two
lines at
higher
field
82.3
170.6
135.8
57.1
52.0
TMS
proton decoupled
δ ppm
0

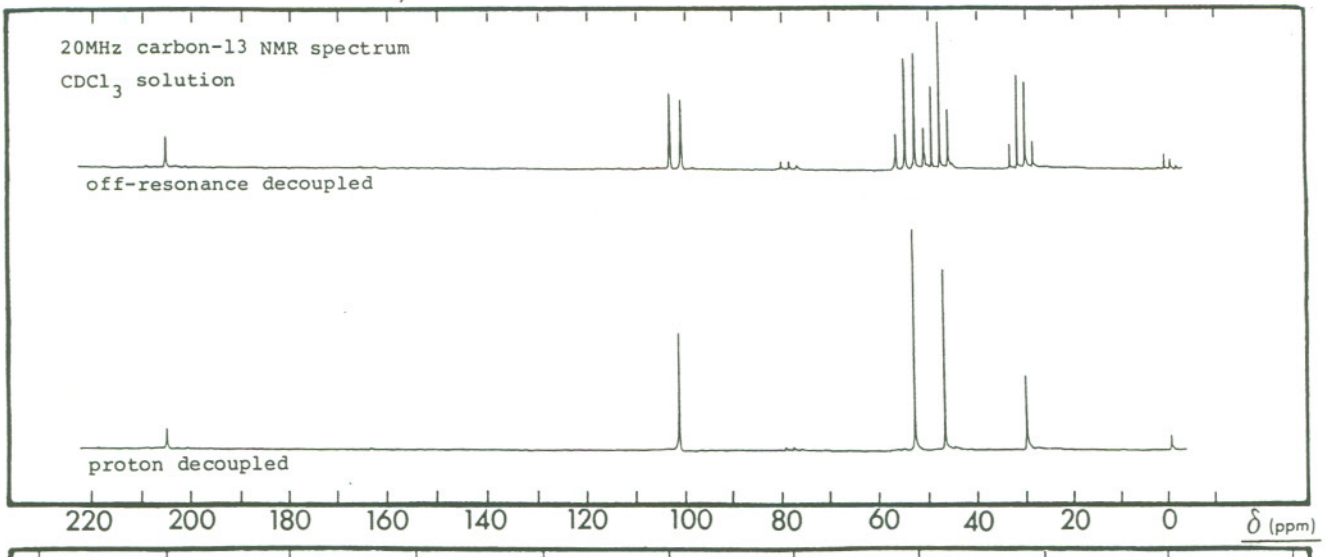

100MHz proton NMR spectrum
CDCl3 solution
10
9
8
7
6
5
4
3
2
1
0
δ ppm

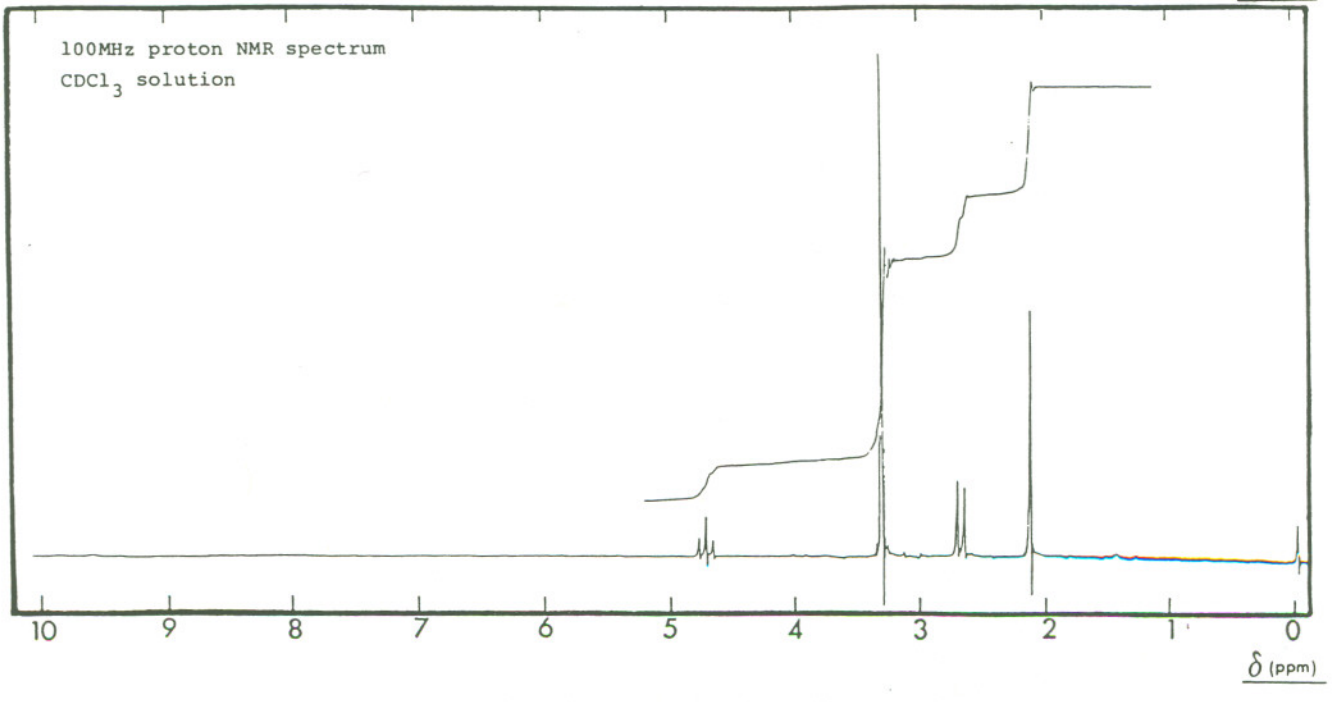
Problem 3


IR spectrum
(liquid film)
1715
4000
3000
2000
1600
1200
800
ν (cm⁻¹)


mass spectrum
43
75
101
M⁺· 132
% of base peak
90
70
50
30
10
20 40 60 80 100 120 140 160 180 200 220 240 260 280 300 320 340
m/e


UV spectrum
no significant
absorption
above 210nm
absorbance
0
0.3
0.6
0.9
1.2
1.5
200
250
300
350
λ (nm)


20MHz carbon-13 NMR spectrum
CDCl3 solution
off-resonance decoupled
proton decoupled
220 200 180 160 140 120 100 80 60 40 20 0
δ (ppm)


100MHz proton NMR spectrum
CDCl3 solution
10 9 8 7 6 5 4 3 2 1 0
δ (ppm)